

ConceptCombo: Assisting Online Video Access with Concept Mapping and Social Commenting Visualizations

Jingxian Liao
jxliao@ucdavis.edu
University of California, Davis
Davis, CA, USA

Mrinalini Singh
mrsingh@ucdavis.edu
University of California, Davis
Davis, CA, USA

Yi-Ting Hung
nctucs.cs08@nycu.edu.tw
National Yang Ming Chiao Tung
University
Hsinchu, Taiwan

Wen-Chieh Lin
wclin@cs.nycu.edu.tw
National Yang Ming Chiao Tung
University
Hsinchu, Taiwan

Hao-Chuan Wang
hciwang@ucdavis.edu
University of California, Davis
Davis, CA, USA

ABSTRACT

As access to educational video resources has been enhanced through search tools, recommender systems, and social channeling, users often encounter challenges in integrating diverse videos systematically, particularly when they possess a limited understanding of the topics at hand (e.g., novice learners). To address this challenge, we present ConceptCombo, a video navigating interface facilitating user explorations of unstructured video collections with conceptual structures and themes of social comments extracted from videos and visualized by the system. To help novices identify a series of quality videos to watch, ConceptCombo aims to deliver a structured overview of the video collection by systematically extracting a concept map from the video content, coupled with automatic summarization of user comments across the videos, as a treemap visualization. With ConceptCombo, novice users may adopt a synergistic approach to video exploration, underpinned by the semantic structure of the video content, social comments, and additional video metadata.

CCS CONCEPTS

• **Human-centered computing** → **Graphical user interfaces**; *Information visualization*.

KEYWORDS

datasets, neural networks, gaze detection, text tagging

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1 INTRODUCTION

As access to video creation tools becomes ubiquitous, people spend a significant portion of their time curating and watching videos to exchange information and knowledge from various topics, including curricular, professional knowledge and academic subjects. Meanwhile, generic video-sharing platforms including YouTube, and specialized online learning platforms such as MOOCs, enable users to access videos from experts and general creators globally. [2]. Extensive social interactions on these platforms also offer opportunities for all users to share their knowledge, experiences, and opinions, which essentially enrich information sources and create a community for social connections [1, 3].

As the volume of videos sparks, there have been substantial technologies and channels designed to promote people to retrieve high-quality videos tailored to their specific needs and preferences. For instance, recommender systems return videos based on personal past watching and search history, video popularity, and content diversity, among others [6, 7]. Another common approach is that people find videos through social sharing in their social networks and online feeds. However, even with the channels above, it remains a challenge for users to find a small number of relevant, high-quality videos to understand specific topics in depth and with breadth systematically. Prior work suggests that users may shirk to find and filter the videos on their own for specific video consumption goals, especially for users with low background knowledge and digital literacy [4]. For example, novices often feel challenged to identify what videos appear to be more useful and what navigation path appears to be more efficient. It is cognitively taxing for people to identify the inner connection of video content and extract trustworthy insights from the comments other audiences left on videos, if they want to fully integrate the content, information, and knowledge distributed across different videos, to guide their video exploration.

To address these challenges, we designed ConceptCombo, an information visualization system that serves as a video navigation interface for topic- and learning-oriented video consumption. It's designed to facilitate video exploration by assembling a concept map visualization of concepts and prerequisite relationships among different concepts embedded in the video content and a treemap visualization representing the common themes extracted

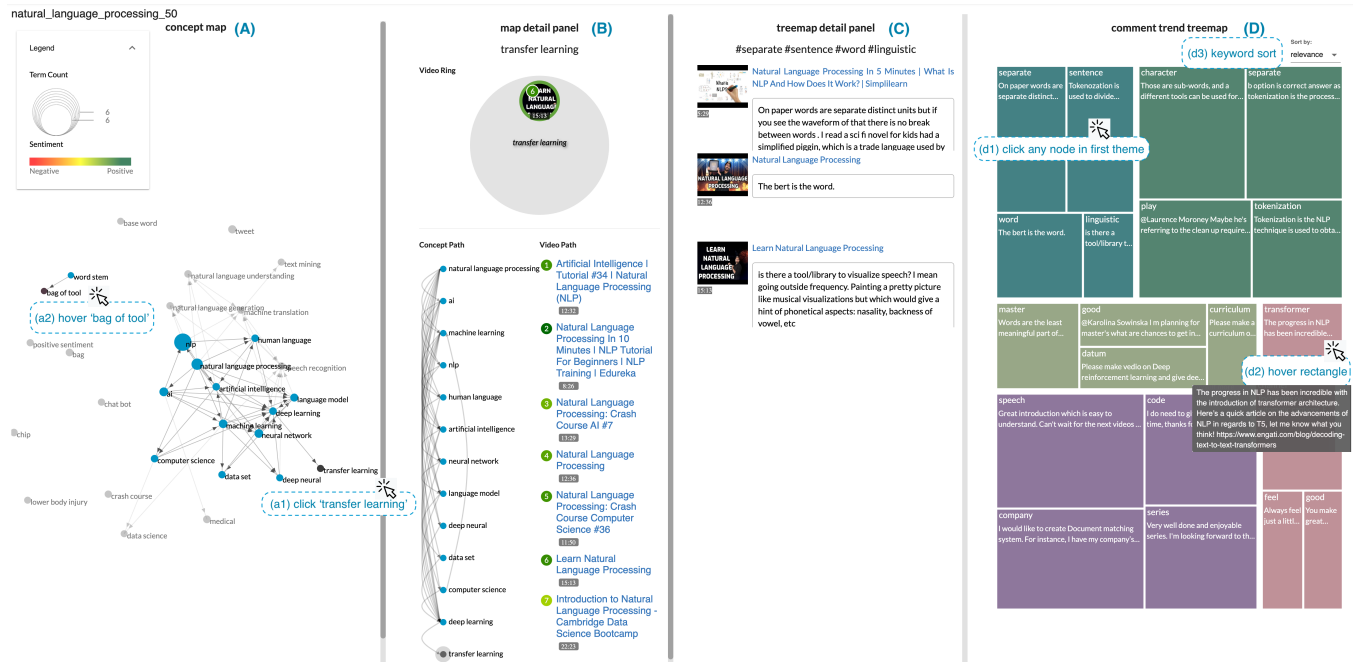


Figure 1: ConceptCombo Overview Page. As a user chooses a video topic ('Natural Language Processing'), ConceptCombo displays a conceptual summary of videos and their comments by concept map (A) and comment treemap (D). As users explore the concept map and click a concept node (a1) in the concept map (A), the map detail panel (B) will provide detailed video recommendations for the selected node ('transfer learning' here), including a ring of recommended videos for the concept, a video sequence and a parallel concept sequence that users may choose to learn from the beginning to the concept. When users check the comment themes, they may hover over a node to see the full content of its pertinent comments or select any theme (d1). Then the treemap detail panel (D) will pop up with several videos with comments in the selected comment themes as recommendations.

from the collection of comments users posted across diverse videos. It displays a comprehensive and structured overview of YouTubers' knowledge structures and topic summaries from peer learners, all extracted automatically from the raw video sources. And It also affords semantic interactions between knowledge structures from the videos and topical summary of comments through the underlying video resources. Users have the flexibility to navigate the video collection by following the guidance of the concept summary via concept map or comment summary via treemap, or even combining both, according to personal interests and specific learning needs.

2 CONCEPTCOMBO SYSTEM

2.1 Interface Design

As users initiate a topic to learn, the video overview is generated from YouTube videos and comments with a computational pipeline that we will introduce later in section 2.2. ConceptCombo visualizes the structured overview with four components, as illustrated in Figure 1: *Concept Map (A)*, *Map Detail Panel (B)*, *Comment Treemap (C)*, and *Treemap Detail Panel (D)*. Concept Map and Comment Treemap encapsulate critical points from video creators' structures and peer learners' focus across videos in different visualization formats, respectively. Users can further delve into more detailed

content and video recommendation via the two central panels from the map and treemap.

- *Comment Trend Treemap (D)* displays a topical overview, summarizing the social discussion themes through topic modeling. Each rectangle in the treemap presents a keyword and displays a snippet of one representative comment for that keyword. Rectangle colors differentiate comment themes, while the area and positioning denote keyword significance. Hovering over a comment keyword rectangle reveals the full content of a pertinent comment (d2 in Figure 1). Simultaneously, Concept Map highlights the knowledge concepts mentioned in the video from which the comment originates. On selecting a keyword rectangle, Treemap Detail Panel lists all videos containing representative comments of the related theme. Users can adjust the sorting of top keywords by relevance to the learning topic or by keyword frequency (d3). Figure 1 illustrates how, upon clicking the keyword "sentence," the primary comment theme is emphasized, with recommended videos and detailed information displayed in Treemap Detail Panel.

- *Treemap Detail Panel (C)* furnishes video recommendations linked to a comment theme corresponding to the clicked keyword. It aids users in identifying a subset of videos whose comments are closely relevant to the chosen theme, which may reflect similarities

in video content, target audience, and focal points. This panel displays various video metadata, including the title, thumbnail, and representative comments from the comment theme. To provide additional concept hints about the videos, hovering over a video emphasizes the concepts mentioned in the video content and comments on both the Concept Map (A) and Comment Treemap (D), respectively.

- *Concept Map (A) and Map Detail Panel (B)* are adopted from the visualization and interaction design of the concept map described in [5]. Concept Map (A) graphically portrays key knowledge concepts and their relationships derived from video transcripts. Central and basic concepts occupy the canvas's center, with advanced concepts surrounding them. Directed links transition from basic to complex concepts, and the size of concept nodes corresponds to their frequency in the videos. Map Detail Panel (B) provides a ring of video recommendations for the clicked concept, a concept path outlining a path from basic prerequisite concepts to the clicked concept, and a video path based on the concept sequence, facilitating users in finding suitable watching sequences. More details can be found in their paper [5]. Similar to interactions in Treemap Detail Panel, hovering over a video thumbnail or title in Map Detail Panel highlights the concepts discussed in the video on Concept Map and the comment keywords in Comment Treemap. Clicking any video metadata in both panels redirects users to the video player to watch the video and its selected comments.

2.2 System Construction

ConceptCombo utilizes an NLP computational pipeline to generate the conceptual structures and themes. Upon receiving a search query, the system collects top videos from the YouTube Data API, retrieving English captions, user comments, and video metadata, such as titles, thumbnails, and like counts. For video overview, the system analyzes all video captions, extracting keyword candidates from transcripts, titles, tags, channel names, and other metadata. It leverages Wikipedia glossary and Google NLP APIs to capture domain-specific keywords and filter irrelevant ones. After concept identification, the system assesses semantic similarity, frequency, and concept co-occurrence among the videos and other materials to discern potential prerequisite relationships between concept pairs. With the concept map generated, it suggests videos for each concept based on the relative importance of the concept in the videos and generates a video sequence recommendation based on map links and video recommendations for each concept. Most of the computational pipeline has been adopted from [5]. For user comment outlines as treemap, ConceptCombo collects the top comments from each video, varying the actual number according to original comment size. It first removes irrelevant comments, such as ads, pure appraisals, and spam. Next, the system uses BERT and LDA embedding and autoencoder models to analyze and create four to six comment topic themes based on their semantic relationship, choosing multiple keywords from each theme for representation. Simultaneously, the system identifies the most connected comments for each keyword and theme, based on their semantic similarity.

3 CONCLUSION

While extensive online video resources provide access to mass knowledge across a range of domains, conventional video navigation often falls short in supporting structured video learning and adaptive navigation to accommodate the knowledge gaps of novice learners. In this demo, we present ConceptCombo, a video navigation system that simultaneously visualizes the concept structure embedded in video content and the integrated view of social comments generated by other users across different videos, coupled with the connections between two overviews. Learners, especially novice learners, may adopt a balanced approach to accessing videos, facilitated by the inherent semantic structure of video resources. In the demo session in CSCW 2023, attendees will have first-hand experience with the visualizations, comparing and benefiting different sources of navigational support to explore the videos of their interests. During the demo session, we will prepare video sets spanning a wide range of topics from generic to professional domains of interest for the attendees to experience.

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